

Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	26 June 2026
Roll No	23A51A05K2
Project Name	Leave Approval System
Maximum Marks	4 Marks

Technical Architecture:

The Deliverable shall include the architectural diagram as below and the information as per the table1 & table 2

Leave Approval System

Mini Architecture Diagram

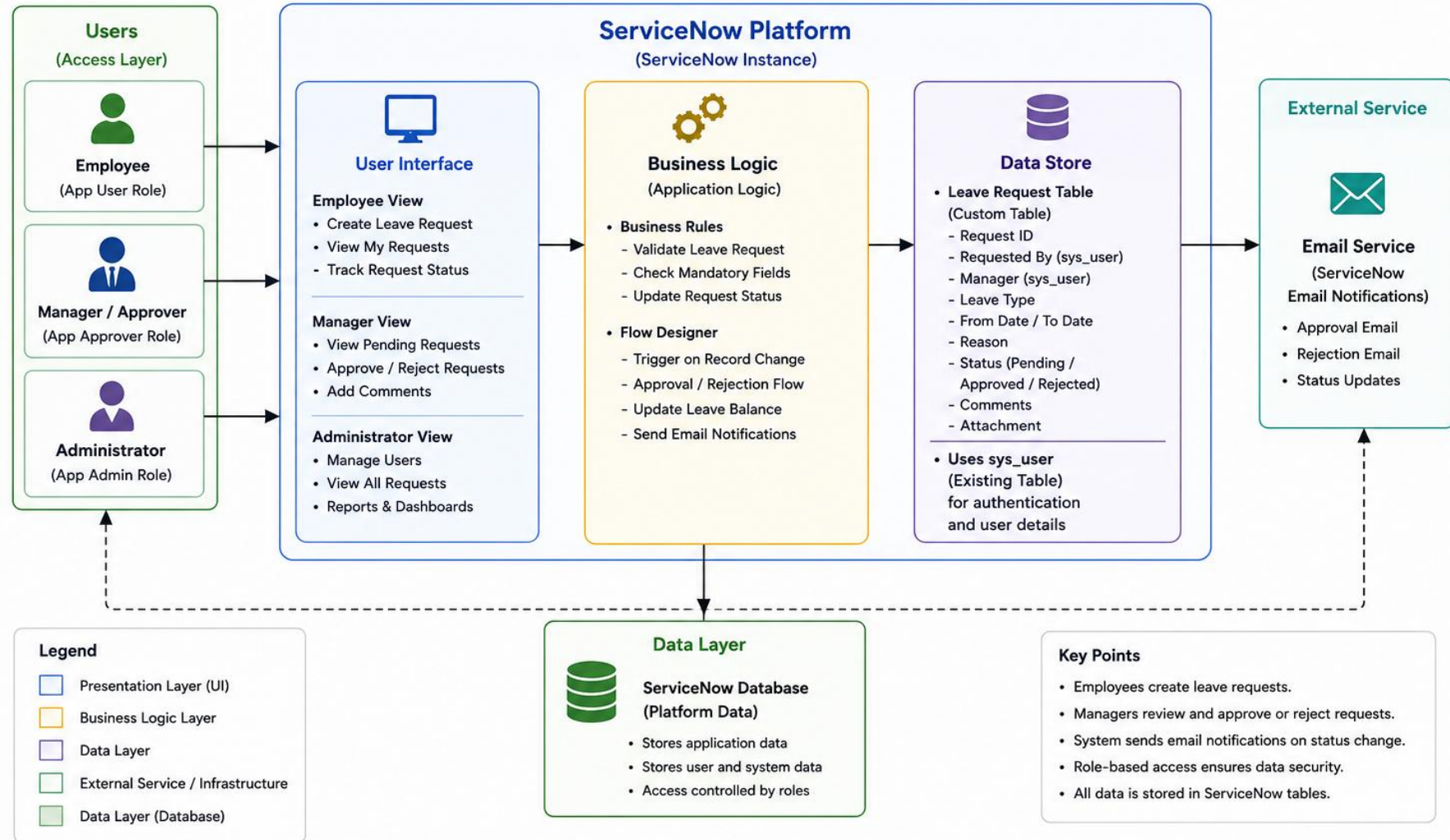


Table 1: Components & Technologies

S.No	Component	Description	Technology
1	User Interface	Employee, Manager, and Administrator views for leave management	ServiceNow Workspace / Forms / Lists
2	Application Logic	Leave request creation, validation, approval, and status management	Business Rules, Client Scripts
3	Workflow Automation	Leave approval workflow, notifications, and record updates	Flow Designer
4	Database	Stores employee details and leave requests	sys_user, Custom Leave Request table
5	Email Service	Sends approval and rejection email notifications	ServiceNow Notifications
6	Security	Role-based access for employees, managers, and administrators	ACLs, Employee Role, Manager Role, Admin Role
7	Infrastructure	ACLs, Employee Role, Manager Role, Admin Role	ServiceNow Cloud Platform

Table 2: Application Characteristics

Characteristic	Description	Technology
Security	Role-based access ensures employees, managers, and administrators access only authorized data.	ACLs, ServiceNow Roles
Scalability	Supports a growing number of employees and leave requests without major changes.	ServiceNow Platform
Availability	The application is available whenever the ServiceNow instance is operational.	ServiceNow Cloud
Performance	Leave requests, approvals, and notifications are processed efficiently.	Flow Designer, Business Rules
Reliability	Leave records and approval history are stored securely with accurate processing.	ServiceNow Database

Maintainability	Forms, workflows, business rules, and notifications can be updated easily without affecting existing data.	ServiceNow Studio, Flow Designer
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